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Technical Requirements

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1. Introductions

1.1. Purpose

The purpose of this project is to oversee the optimization of growing environment of fungi, where the project and its resources can be utilized for other purposes with small changes and optimization

1.2. Target Audience and Application

The target audience is farms, private persons and businesses

1.3. Definitions

IDE - Integrated Development Environment

VSCoDe - Visual Studio Code

VOC - Volatile Organic Compounds

eCO₂ - equivalent CO₂

2. Overall Description

2.1 System Overview

Key components:

- Adafruit Feather RP2040
- Adafruit SGP30 Air Quality Sensor Breakout - VOC and eCO2
- Adafruit AHT20 - Temperature & Humidity Sensor Breakout Board
- Piezo Buzzer
- Protoboard
- 1 Button
- Quad alpha numeric display

2.2 System Limitations

- Limited processing and memory on the Adafruit Feather RP2040
- Limitations such as sensor range and calibration accuracy

3. Functional Requirements

3.1. Data Collection

FR1. The system shall collect data on VOC and eCO₂ using the Adafruit SGP30 sensor once per second (1 Hz). The raw sensor values should be stored in the RAM.

FR2. The system shall measure temperature and humidity using the Adafruit AHT20 sensor every two seconds (0.5 Hz)

3.2. Data Processing and Analysis

FR4. The system shall compare the sensor readings against predefined threshold values for shitake mushrooms.

FR5. TVOC $\geq$ 500 ppb $\rightarrow$ Level-1 alarm

eCO₂ $\geq$ 1000 ppm $\rightarrow$ Level-1 alarm

eCO₂ $\geq$ 2000 ppm $\rightarrow$ Level-2 alarm

Temperature $< 18^{\circ}\text{C}$ or $> 26^{\circ}\text{C}$ $\rightarrow$ Temp alarm

Humidity $< 30\% \text{RH}$ or $> 60\% \text{RH}$ $\rightarrow$ Humidity alarm

FR6. Level-1: yellow neopixel + one 200 ms beep every 5 s

Level-2: red neopixel + three 100 ms beeps at 2 Hz every 10 s

Temp/Humidity: Blue (low), red (high) neopixel blink

FR7. Pressing the encoder button while an alarm is active shall silence the buzzer but leave the neopixel on until the reading returns below threshold

FR8. If two alarms set off at the same time it shall switch both lights back and forth in a 5 second interval until it comes back to within its threshold. The alpha numeric display shall also display the both v

3.3. User Interface, Display and alarms

FR5. The system shall display current sensor readings on a Quad alphanumeric display every two seconds

FR6. Data shall be shown in 4 character blocks:

- Temperature
- Humidity
- TVOC
- eCO₂
- The display shall cycle through the parameters every 10 seconds in the order
T → H → V → C.

FR8. The piezo buzzer should be used when initiating the alarm.

FR9. all limits are stored in flash

4. Non Functional Requirements

4.1. Performance

NFR1. The system shall update and display values of the measurements every 2 seconds.

4.2. Usability

NFR3. The usability of the system shall be simple and easy for users to understand.

NFR4. The systems hardware such as LCD screen, shall be easy to understand and analyze for the user

4.3. Reliability

NFR6. The system shall run for 8h continuously without crash

NFR7. If any of the sensors fail to reply for 3 consecutive polls, it shall write an error message in the serial log and restart.

5. System Interfaces

5.1 Hardware Interfaces

5.1.1 I2C Communication

- SGP30 (CO2/eVOC)
- AHT20 (Temperature/Humidity)

5.1.2 PWM

- Piezo Buzzer

5.1.3 GPIO

- 1 Button

5.2 Software Interfaces

5.2.1 Libraries

- “adafruit_gp30” for CO2/eVOC readings
- “adafruit_aht20” for temperature/humidity data
- “pwmio” for PWM control
- “board” for I2C control

6. Design Constraints and Assumptions

6.1. Technical Constraints

Hardware Limitations: The Adafruit Feather RP2040 is the core microcontroller, which has limited processing power and memory. Sensors such as the Adafruit SGP30 (VOC & eCO₂) and AHT20 (temperature & humidity) are constrained by range and calibration accuracy. These limitations impose a cap on the complexity of the codebase, sensor sampling rate, and processing logic.

Component Integration: The system must integrate at least two different sensors (already fulfilled by SGP30 and AHT20). Must provide feedback/output using the display, buzzer and neopixel—the project includes a Quad alphanumeric display and a Piezo buzzer.

State Machine Design: The system logic must be structured as a state machine to manage different operational modes (e.g., idle, measuring, alerting, error handling). This adds a constraint on software architecture.

6.2. Environmental Usage Assumptions

Use in controlled environments (e.g., farms, greenhouses). Tolerance to typical temperature and humidity variations. Stable power supply or battery operation constraints.

7. Testing Requirements and Verifications

7.1 Functional Testing

Sensor Validation: Ensure the VOC, eCO₂, temperature, and humidity readings are collected and processed accurately. Validate thresholds and correct triggering of alarms or feedback mechanisms (buzzer/display).

State Transitions: Test all transitions between states in the state machine: From idle to active, active to alert, alert to reset, etc. Simulate scenarios like sensor faults or environmental anomalies to confirm appropriate responses.

User Interaction: Validate correct input recognition from buttons or any other interactive components. Ensure the display shows accurate and updated data.

7.2. Non Functional Testing

Performance: Ensure that sensor values are updated and displayed every 2 seconds. Check for any latency or lag in data collection or interface feedback.

Usability: Test the interface for ease of use—ensure users can intuitively read outputs and understand system states. Verify that display messages and buzzer alerts are clear and meaningful.

Reliability: Simulate errors (e.g., sensor disconnection, power loss) and verify that the system remains usable or fails gracefully.

Sensor disconnect: unplug the SGP30 sensor, the device should answer with the message “SENSOR ERROR: disconnected” in the serial log.

Power loss: cycle power, the device should run back to its normal operation and retain stored thresholds and values

Maintainability: Check that the hardware setup and codebase are structured and documented for easy maintenance and extensibility